

CP2 Lab Control Seed Viability

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Analysis overview

Each control seed is treated as one replicate. Seed viability is therefore a binary response: 1 = viable and 0 = not viable.

The analysis is repeated using three definitions of viability:

1. Category 1 only
2. Categories 1 and 2
3. Categories 1, 2, and 3

Because the response is binary, the binomial logistic regression is treated as the primary model. A linear model and ANOVA are also included as requested.

Definition 1: Category 1 only

Population summary

Population	Sample size	Viability (%)	Lower 95% CI (%)	Upper 95% CI (%)
TM2	30	46.7	28.8	64.5
BH	30	43.3	25.6	61.1
SC	29	58.6	40.7	76.5
IH	30	43.3	25.6	61.1
LV2	29	48.3	30.1	66.5
WL1	29	3.4	0.0	10.1
DPR	28	32.1	14.8	49.4
WL2	28	28.6	11.8	45.3
WL3	30	10.0	0.0	20.7
YOSE1	30	33.3	16.5	50.2
SQ3	8	25.0	0.0	55.0

Figure

The small jittered points represent individual seeds. The larger points show the mean viability proportion for each population, and the vertical bars show approximate 95% confidence intervals.

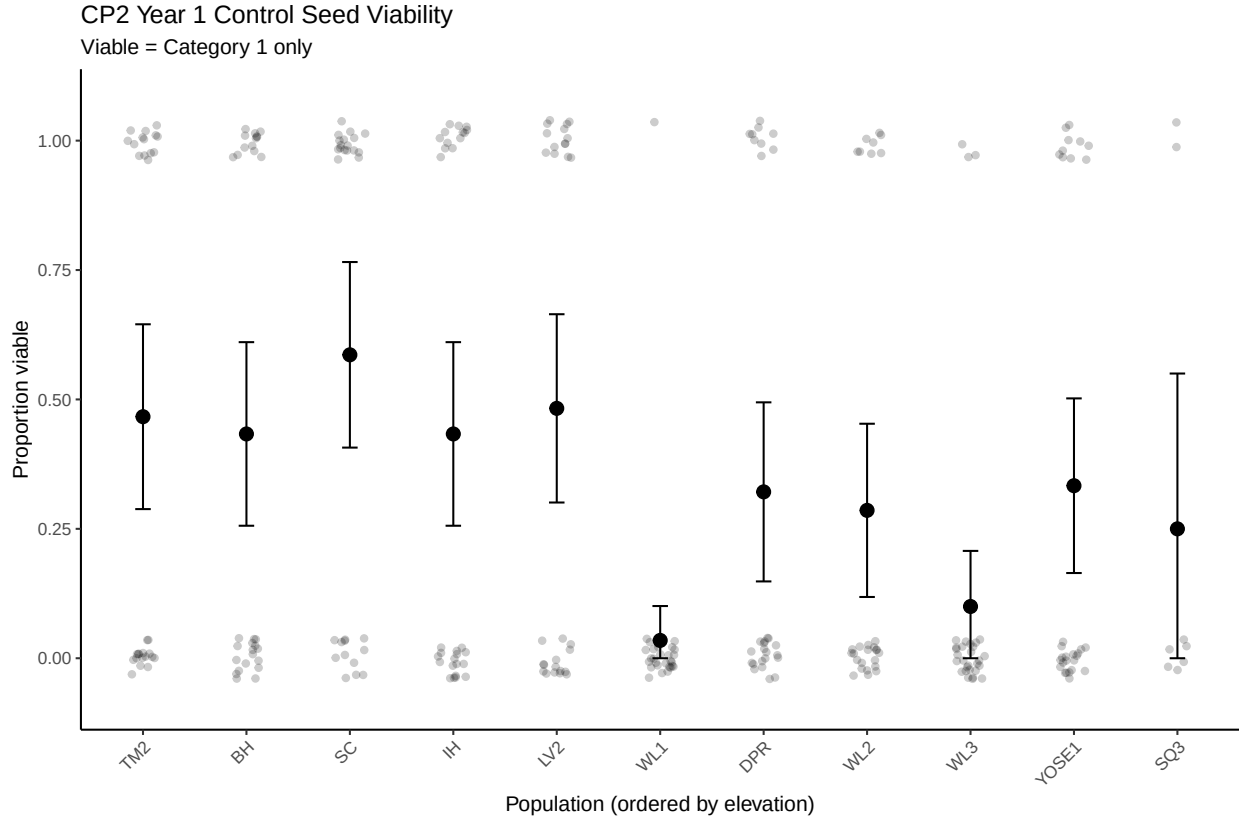


Figure 1: Control seed viability when Categories 1 is considered viable.

Linear model and ANOVA

term	df	sumsq	meansq	statistic	p.value
Pop	10	7.9370	0.7937	3.8279	0.0001
Residuals	290	60.1295	0.2073		

Binomial logistic regression

term	df	deviance	df.residual	residual.deviance	p.value
NULL			300	388.0648	
Pop	10	40.9367	290	347.1281	0

Definition 2: Categories 1 and 2

Population summary

Population	Sample size	Viability (%)	Lower 95% CI (%)	Upper 95% CI (%)
TM2	30	60.0	42.5	77.5
BH	30	63.3	46.1	80.6
SC	29	62.1	44.4	79.7
IH	30	70.0	53.6	86.4
LV2	29	100.0	100.0	100.0
WL1	29	6.9	0.0	16.1
DPR	28	57.1	38.8	75.5
WL2	28	35.7	18.0	53.5
WL3	30	20.0	5.7	34.3
YOSE1	30	40.0	22.5	57.5
SQ3	8	50.0	15.4	84.6

Figure

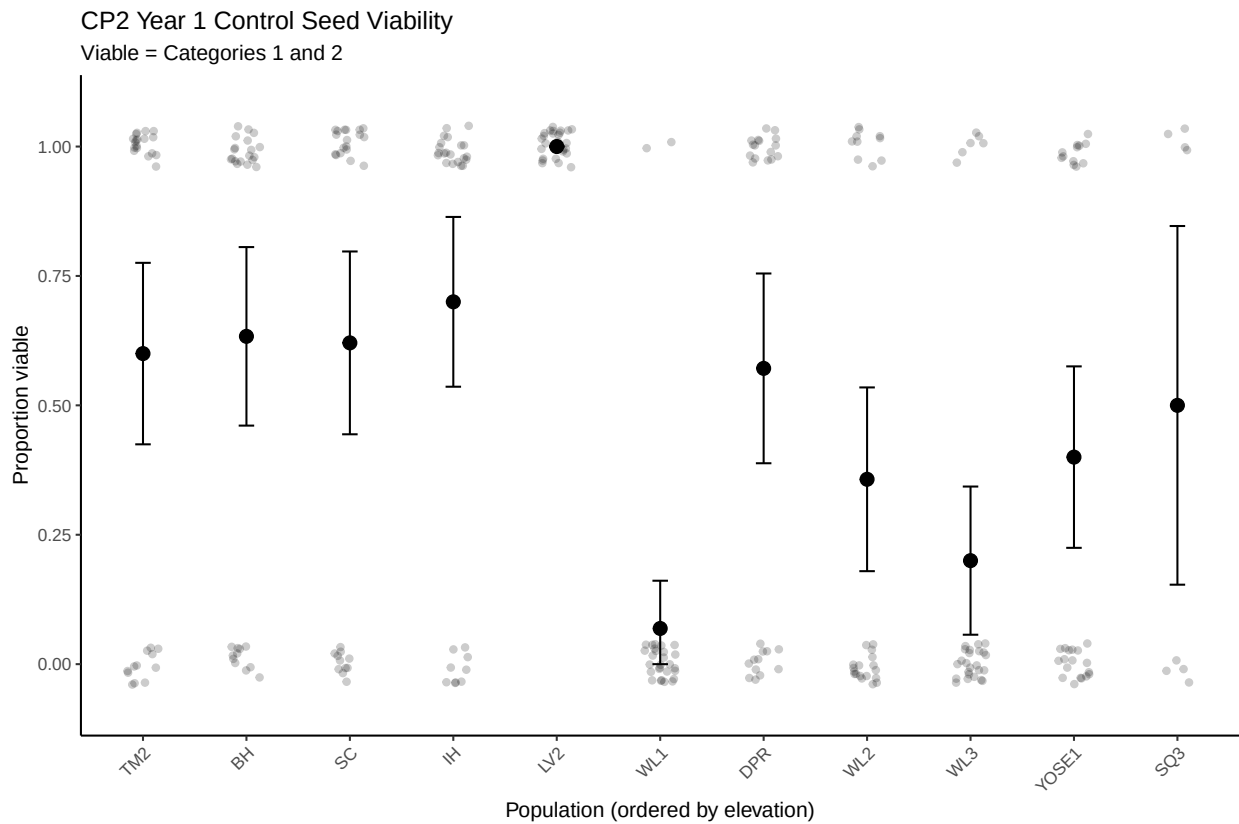


Figure 2: Control seed viability when Categories 1 and 2 are considered viable.

Linear model and ANOVA

term	df	sumsq	meansq	statistic	p.value
Pop	10	18.7407	1.8741	9.629	0
Residuals	290	56.4420	0.1946		

Binomial logistic regression

term	df	deviance	df.residual	residual.deviance	p.value
NULL			300	417.0055	
Pop	10	91.2556	290	325.7499	0

Definition 3: Categories 1, 2, and 3

Population summary

Population	Sample size	Viability (%)	Lower 95% CI (%)	Upper 95% CI (%)
TM2	30	93.3	84.4	100.0
BH	30	80.0	65.7	94.3
SC	29	82.8	69.0	96.5
IH	30	86.7	74.5	98.8
LV2	29	100.0	100.0	100.0
WL1	29	55.2	37.1	73.3
DPR	28	82.1	68.0	96.3
WL2	28	46.4	28.0	64.9
WL3	30	40.0	22.5	57.5
YOSE1	30	56.7	38.9	74.4
SQ3	8	50.0	15.4	84.6

Figure

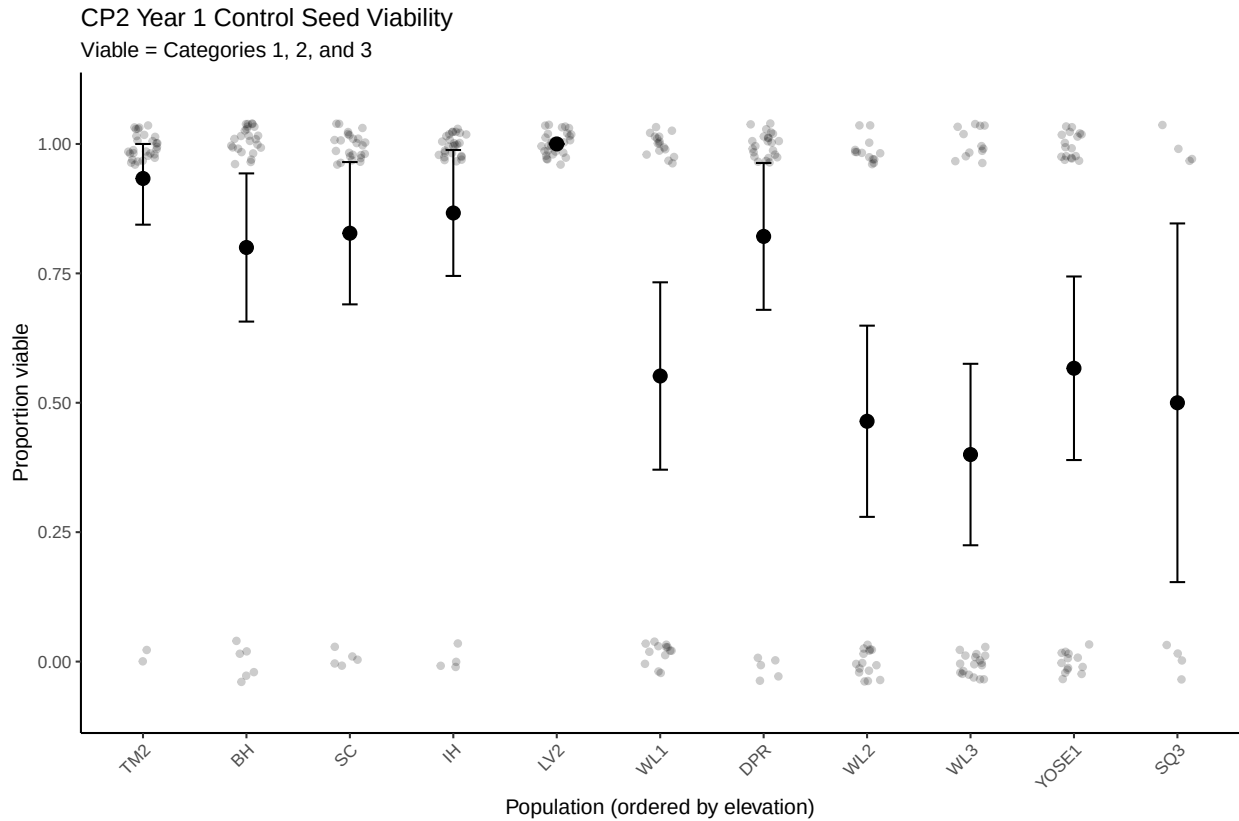


Figure 3: Control seed viability when Categories 1, 2, and 3 are considered viable.

Linear model and ANOVA

term	df	sumsq	meansq	statistic	p.value
Pop	10	11.9149	1.1915	7.0399	0
Residuals	290	49.0818	0.1692		

Binomial logistic regression

term	df	deviance	df.residual	residual.deviance	p.value
NULL			300	358.3094	
Pop	10	66.0006	290	292.3088	0

Notes

- The overall ANOVA and likelihood-ratio tests evaluate whether viability differs among populations.
- SQ3 has a smaller sample size than the other populations, so its estimate is less precise.

- The confidence intervals shown in the figures are descriptive normal-approximation intervals.
- The binomial logistic regression is the preferred inferential analysis for the binary seed-level response.